

Data Alchemy: Extracting Financial Relevance through the Lens of Solar Geomagnetic Activity

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Abstract

Are there unconventional external factors that could influence financial markets even though they are rarely considered? We explore one candidate, geomagnetic activity (“solar storms”), measured by the Kp and ap indices published by GFZ Potsdam [1]. Because the form of a possible solar–market relationship is unknown, we test it with two deliberately different methods: an unsupervised Non-negative Matrix Factorization (NMF) of yearly NASDAQ price trajectories (annual scale), and a supervised 1D convolutional neural network (CNN) that tries to detect market shocks from 14-day windows of Kp (daily scale) on five assets, including Iridium and Delta Airlines, which we consider a priori exposed to space weather.

Neither method provides evidence of a solar effect. The NMF, in the form reported here, decomposes NASDAQ prices alone and is not a test of solar influence. The CNN collapses to a near-constant prediction on all five assets, and the class it outputs is the majority class of each test set; its results are therefore explained by shock frequencies and carry no information about Kp. We present this as an exploratory negative result, explain why it does not rule out a weak effect, and list what a properly powered test would require.

1 Introduction

Traditional data streams (financial statements, news) are heavily exploited, so any residual edge in them is likely small. Extreme physical phenomena are an unconventional alternative: they are exogenous to the market and may offer a “pure” yet noisy signal. We focus on geomagnetic activity, through the Kp index (GFZ Potsdam), for two reasons: it is a non-intuitive external factor, and long, usable daily records are freely available.

Geomagnetic storms can disturb satellite operations and polar flight routes. We therefore consider Iridium (IRDM), a satellite operator, and Delta Airlines (DAL), an airline relying on polar-route stability, as a priori exposed assets. Our two working hypotheses are:

- **H1 (annual scale):** yearly market trajectories and yearly geomagnetic activity share latent temporal patterns.

- **H2 (daily scale):** specific sequences of Kp values over the preceding 14 days carry information about the occurrence of a market shock.

We do not expect large effects: markets are competitive and the prior is low. The aim is a first, honest feasibility test.

Why two methods. Without prior knowledge of how a solar signal would enter prices, a single model would be an arbitrary choice. The two methods differ in scale, supervision and assumptions, and each covers a blind spot of the other. NMF is unsupervised, makes few assumptions and is interpretable; at the annual scale it is suited to slow effects such as the ~ 11 -year solar cycle. The 1D-CNN is supervised, non-linear, and targets short-term signatures of a few days. If both fail to reveal a link at their respective scales, this is converging (though not conclusive) evidence; if one had succeeded, the other would have served as a cross-check. Reporting such a design, even with a null outcome, delimits what is *not* detectable with these data and tools and leaves a reusable pipeline.

2 Methodology and Techniques

2.1 Data and reproducibility

The Kp and ap indices come from GFZ Potsdam and are freely available at https://www-app3.gfz-potsdam.de/kp_index/Kp_ap_since_1932.txt [1, 2]. They cover 1932 to 2024; Kp is a 3-hourly, quasi-logarithmic index from 0 to 9 and ap is its linear counterpart. Daily prices of the financial assets were downloaded from a public provider (not recorded).

The analysis code was not archived. The figures and numbers below are those obtained during the hackathon and are reported as an exploratory record, not as a reproducible benchmark. Where a detail is not documented (definition of a “shock”, train/test split, hyper-parameters), we say so instead of guessing.

2.2 Non-negative Matrix Factorization (NMF): Pattern Extraction

The first phase of our “Alchemy Engine” uses **Non-negative Matrix Factorization (NMF)**. Unlike standard PCA, NMF imposes positivity constraints, which makes the components additive and easier to interpret. It remains a *linear* low-rank model: the constraint does not make it non-linear.

The intended use is to extract components for both Kp and asset prices over one year and to compare them by plotting them on the same scale. This comparison is not reported here (see Section 3.1).

NMF factors an $N \times p$ data matrix $\mathbf{X}$ into two matrices with non-negative entries, $\mathbf{W}$ ($N \times r$) and $\mathbf{H}$ ($r \times p$):

$$\mathbf{X}_{N \times p} \approx \mathbf{W}_{N \times r} \mathbf{H}_{r \times p},$$

subject to

- $w_{i,j} \geq 0$ and $h_{k,j} \geq 0$ (non-negativity);
- $r < \min(N, p)$ (rank reduction).

Since $r < p$, the product $\mathbf{WH}$ is a low-dimensional approximation of $\mathbf{X}$.

NMF requires a non-negative input. To prevent features on different scales from dominating the cost function, we normalise the data to fall between 0 and 1.

For the Kp, $\mathbf{X}$ is the matrix whose rows are yearly series of daily Kp ($p = 365$), with $N \approx 92$ yearly samples (1932–2024). The aim is to separate a geomagnetic “background” from episodic storm activity. Results for this matrix are not part of this report.

The decomposition minimises the Euclidean (Frobenius) distance between $\mathbf{X}$ and $\mathbf{WH}$ with Lee & Seung’s multiplicative update rules [3], an iterative scheme that refines $\mathbf{W}$ and $\mathbf{H}$ over successive cycles. The objective is non-convex, so the solution depends on the random initialisation.

2.3 Convolutional Neural Network (1D-CNN): Signature Detection

To capture temporal dynamics, we implement a 1D Convolutional Neural Network. Its core is a *kernel* of size 3, acting as a sliding window able to recognise sequential patterns (e.g., a “calm before the storm” followed by a sharp increase).

- **Input:** 14-day sliding windows of Kp-Index activity.
- **Architecture:** Conv1D layers for feature extraction, MaxPooling for noise reduction, and heavy Dropout (0.5) to limit overfitting.
- **Loss function:** Weighted binary cross-entropy, intended to handle class imbalance between calm days and market shocks.

The precise definition of a market shock, the prediction horizon, the train/test split and the remaining hyper-parameters were not archived and are therefore not reported.

3 Analysis and Results

3.1 NMF

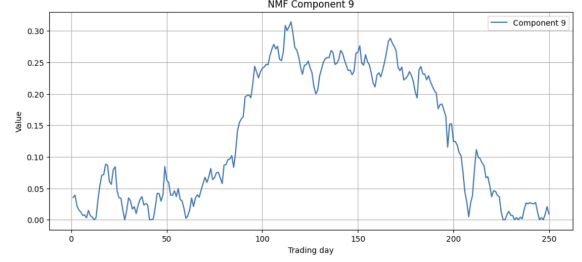


Figure 1: NMF component 9 of the yearly NASDAQ price matrix (normalised values, position of the trading day within the year).

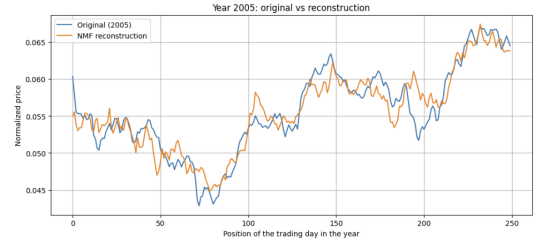


Figure 2: Original versus NMF-reconstructed normalised NASDAQ path for the year 2005 (in-sample).

The financial data are the NASDAQ Composite index over 2004–2024. We downloaded historical market data and defined a daily price as the average of the daily Open and Close. After merging with the solar data on common dates, we obtained 5,033 daily observations of three variables: the market price, the Kp index and the ap index. The price series was then reorganised into a yearly matrix, one row per year and one column per trading-day position within the year, giving a 20×250 matrix.

NMF was applied to the normalised price matrix. The decomposition returned a matrix of yearly weights and a matrix of components, with a reconstruction error of about 0.213 (the exact error metric was not recorded). Figure 2 shows the reconstruction for 2005; the reconstructions for 2010 and 2015 look similar. Figure 1 shows one of the components. The number of components r was not recorded; the index 9 in Figure 1 implies $r \geq 9$.

Table 1: CNN results on the test set of each asset. *Shock rate* and *F1 base* are derived from the confusion matrix: $p = (FN + TP)/n$, and $F1 \text{ base} = 2p/(1 + p)$ is the F1 of the trivial classifier that always predicts “shock”.

Ticker	TN	FP	FN	TP	F1	Shock rate	F1 base
DAL	2	20	0	32	0.76	59%	0.74
IRDM	0	21	0	30	0.74	59%	0.74
H.TO	23	0	5	0	0.00	18%	0.30
^GSPC	54	0	16	0	0.00	23%	0.37
LMT	49	0	21	0	0.00	30%	0.46

DAL = Delta Airlines, IRDM = Iridium, H.TO = Hydro One, ^GSPC = S&P 500, LMT = Lockheed Martin. TN = true negative, FP = false positive (false alarm), FN = false negative (missed shock), TP = true positive. F1 is computed for the “shock” class. Test sets contain between 28 and 70 samples.

What this does and does not show. It shows that yearly NASDAQ paths admit a compact non-negative representation. It does *not* show a solar influence, for four reasons. (i) The price matrix was decomposed alone and never confronted with Kp components or weights. (ii) With only 20 rows and at least 9 components, a good in-sample reconstruction is expected, and no null reference (for instance random-walk surrogates) was computed. (iii) The values in Figure 2 (≈ 0.05 in 2005) suggest a normalisation common to all years, so that rows differ mostly by overall level, i.e. by the long-run growth of the index, and components partly encode the trend rather than recurring shapes. (iv) We therefore make no claim that the components are “meaningful structural variations”.

Our answer to the research question at this scale is thus a null one: this framework provides no evidence that solar storms influence the NASDAQ at the annual level. This is not proof of absence. After aggregation, the analysis rests on 20 yearly observations, a very small sample, and the effective number of independent solar cycles covered is only about two. Annual summaries can also smooth out short-term reactions. If an effect exists, it may be weaker, indirect, non-linear, or visible only at a finer time scale, which motivates the daily-scale CNN below.

3.2 1D

CNN

3.2.1 Results

3.2.2 Reading the Performance Metrics

Accuracy is misleading when classes are unbalanced, so we look at the confusion matrix and the F1-score of the “shock” class (Table 1):

- **TP & FN** define the *recall*: a high TP and low FN mean that the shocks were caught.
- **FP** are false alarms. In risk management, a higher FP rate can be acceptable to avoid missing tail risks.

- **F1** is the harmonic mean of precision and recall. It must be read against the base rate: a classifier that always predicts “shock” obtains $2p/(1 + p)$, i.e. 0.74 when shocks are 59% of the samples. An F1 above 0.5 is therefore *not* evidence of skill, and an F1 of 0.00 is what a classifier that always predicts “calm” obtains.

Note that shocks are the minority class for H.TO, ^GSPC and LMT (18–30%) but the *majority* class for DAL and IRDM ($\approx 59\%$).

3.2.3 Interpretation

of

Results

Three observations can be read directly from Table 1.

- (a) *Constant predictions.* On every asset the network outputs (almost) a single class: nearly always “shock” for DAL and IRDM, always “calm” for H.TO, ^GSPC and LMT. A network that had extracted information from Kp windows would vary its output over time, in a way not dictated by the base rate.
- (b) *The constant is the majority class of the test set.* In all five rows the predicted class coincides with the most frequent true class.

- (c) *F1 tracks the base rate.* For IRDM the F1 (0.74) equals that of the trivial “always shock” classifier, and for DAL it exceeds it by 0.02, which corresponds to two true negatives. The zeros elsewhere are the value obtained by “always calm”.

We therefore do not interpret this degenerate behaviour as evidence of a solar effect, nor as evidence against one. The contrast between DAL/IRDM (“shock” predicted) and the other assets (“calm” predicted) reflects the frequency of shocks in each test set, which depends on the asset, on the shock definition and on the test window, and is independent of Kp. The grouping “exposed versus insulated” is not supported by the table either: Hydro One (power grid) and Lockheed Martin (space and defence) are also plausibly exposed to space weather, yet behave like the S&P 500. What the experiment shows is that, in this setup, the CNN did not learn any Kp signal that beats the class prior.

This is a weak test: the test sets are small (28–70 samples), no control experiment was run (e.g. shuffled Kp, or a model without Kp), and no repetition or confidence interval is available. We can neither confirm nor exclude a weak short-term effect.

4 Conclusion

We asked whether geomagnetic activity carries information about financial markets, and tested it with two methods that differ in time scale and assumptions. Neither provides evidence of an effect: the NMF was applied to NASDAQ prices only and does not test a solar link, and the CNN collapsed to the class prior on all five assets. The result is negative and exploratory, and it does not establish the absence of an effect: the samples are small, the annual scale is coarse and the test sets are tiny.

The value of this work lies elsewhere. (i) It states hypotheses tied to physical exposure channels (satellites, polar routes).

(ii) Its two-scale design lets each method cover the blind spot of the other, so a null result at both scales is more informative than a null result at one. (iii) It highlights pitfalls of this kind of analysis that are easy to miss: F1 must be compared with a trivial baseline, class frequencies differ across assets, and degenerate predictions must not be read as findings. (iv) It yields a precise list of what a decisive test requires (next section).

5 Vision and Future Work

Extending the approach to other space-weather variables (cosmic-ray flux, ionospheric activity) is natural, but only after the following steps:

- **Baselines and controls:** majority-class and no-Kp models, shuffled-Kp and permutation tests, and metrics reported against the base rate (balanced accuracy, MCC, PR-AUC).
- **Sound evaluation:** chronological (walk-forward) splits with a gap at least as long as the 14-day window to avoid overlap leakage, several random seeds and confidence intervals.
- **Shock definition:** one explicit definition, fixed a priori and comparable across assets (e.g. relative to each asset’s own volatility), with a correction for multiple testing across assets.
- **Event studies:** focus on the rare severe storms rather than on the full continuous Kp series.
- **NMF:** decompose Kp and price matrices side by side and test the association of their yearly weights against surrogates; use returns rather than price levels; choose r by cross-validation and compare the reconstruction with a random-walk null.
- **Solar cycle:** account for the ~ 11 -year cycle, since 20 years of data contain only about two cycles and shared regimes could create spurious associations.

References

- [1] J. Matzka, C. Stolle, Y. Yamazaki, O. Bronkalla and A. Morschhauser, “The geomagnetic Kp index and derived indices of geomagnetic activity,” *Space Weather*, 19, e2020SW002641, 2021.
- [2] GFZ German Research Centre for Geosciences, Kp and ap indices since 1932. https://www-app3.gfz-potsdam.de/kp_index/Kp_ap_since_1932.txt
- [3] D. D. Lee and H. S. Seung, “Algorithms for non-negative matrix factorization,” *Advances in Neural Information Processing Systems 13*, 2001.